

WEEKLY SCHEDULE WINTER 2026

(also known as Course Section Information)

CST8506 – Advanced Machine Learning

Section 100 – Synchronous

Artificial Intelligence Software Development

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Learning Resources

Required Resources

Required Textbook:

- We will be using the same textbook from level 1 – CST8502 course.

Hardware and Software:

- RapidMiner
- Python 3.7 or up
- Visualization tools
 - Power BI
 - Tableau

Recommended Resources:

- Kelleher, J. D., D'Arcy, A., & Namee, M. B. (2020). *Fundamentals of machine learning for Predictive Data Analytics: Algorithms, worked examples, and case studies, Second Edition*. Cambridge, MA: The MIT Press. ISBN: 978-0-262-04469-1
- Cady, F. (2017). *The data science handbook*. John Wiley & Sons, Inc.
- Bishop, C. M. (2016). *Pattern Recognition and Machine Learning*. Springer New York.
- Additional books, articles, websites, etc. will be given as required.

Evaluation Breakdown

Assessment	Value	CLRs
Assignments (2 assignments, 10% each)	20%	1-5
Labs with demonstrations	20%	1-5
Final Project with presentation and report	20%	1-5
Midterm	15%	1-4
Final Exam	25%	1-5

Learning Schedule *(subject to change with notification)*

Date	Weekly Theme and Learning Outcomes	Learning Activities	Assessments (%)	Resources	CLRs
Week 1 Jan 14	Data Preprocessing - Feature selection and extraction - Curse of dimensionality and dimensionality reduction - PCA, LDA	- Lecture - Hybrid			1
Week 2 Jan 21	Support Vector Machines	- Lab - Lecture - Hybrid	Lab 1 Due: Jan 23	Ch 8	1-4
Week 3 Jan 28	Deep Learning and Neural Networks CNN	- Lab - Lecture - Hybrid	Lab 1 Demo: Jan 26 Lab 2 (4%) Due: Jan 30	Ch 7.4.7	1-4
Week 4 Feb 4	Deep Learning and Neural Networks - RNN - Time Series Analysis	- Lab - Lecture - Hybrid	Lab 2 Demo: Feb 2 Lab 3 (4%) Due: Feb 6	Ch 6	1-4
Week 5 Feb 11	Bayes Algorithm, Naïve Bayes	- Lab - Lecture - Hybrid - Assignment	Lab 3 Demo: Feb 9 Assignment 1 (10%) Due on Feb 13	Ch 10	1-4
Week 6 Feb 18	Clustering Algorithms	- Lab - Lecture - Hybrid	Assignment 1 Demo: Feb 23 Lab 4 (4%) Due: Feb 20		1-4
Week 7 Feb 25	Midterm	- Lab - Midterm - Hybrid	Midterm (15%) Work period in lab		1-4
Week 8	BREAK	BREAK	BREAK	BREAK	
Week 9 Mar 11	Class Imbalance Problems and Outlier Detection One-class SVM	- Lab - Lecture - Hybrid	Lab 4 Demo: Mar 9 Lab 5 (4%) Due: Mar 13		1-4
Week 10 Mar 18	Classifier Fusion	- Lab - Lecture - Hybrid - Assignment	Lab 5 Demo: Mar 16 Assignment 2 (10%) Due on Mar 20		5
Week 11 Mar 25	Power BI	- Lab - Lecture - Hybrid - Project	Assignment 2 Demo: Mar 23 Project Part 1 Due: Mar 27		5
Week 12 Apr 1	Review	- Lab - Lecture - Hybrid - Project	Project Part 1 Demo: Mar 30 Final Project Due: April 2		1-5
Week 13 Apr 6-8	Project Presentations	- Hybrid - Project	Project (20%) Apr 6-15		1-5
Week 14 Apr 13-15	Project Presentations	- Hybrid - Project	Project (20%) Apr 6-15		1-5
Week 15 Apr 18-25	Final Exam (25%)				

Other Important Information

- For hybrid content, please check Brightspace → Weekly Learning Materials → Hybrid Materials.
- For lectures and labs in-person or online, everyone is supposed to come/join at the start of the class. Instructions will not be repeated.
- For labs, students will be marked absent 30 minutes after the start of the period.
- Labs and Assignments must be submitted according to the Submission Standards published on Brightspace beside the Course Outline.
- Policy AA21 Deferred Evaluation will be applied to all missed evaluations. In the case of missed Lab/Assignment demonstrations due to illness or other valid reasons (See Policy AA21 Deferred Evaluation for valid reasons) may be performed at the next lab period without special arrangement if the demonstrated work has already been submitted by the submission due date.
- For submissions, students who fall under any exceptional circumstances (as outlined in AA21) are expected to submit the work completed to date and address any necessary accommodation or adjustments afterward through the appropriate channels.
- For labs and assignments, students must submit both the corresponding answer document, and all required supporting documents (e.g., Python code files, R Markdown files). Failure to comply with either requirement will result in a grade of zero.
- Labs and assignments are due on Fridays at 11:59 PM, but the work can be submitted until Sunday 11:59 PM of that week without any penalty. This extended due date is set to accommodate all learners.
- For Brightspace submissions (labs, assignments and project), files should NOT be in zipped format. Zipped files will not be graded.
- Labs, assignments and part 1 of the project must be demonstrated during the scheduled lab period (the following week after the due date) to get grades.
- It is the student's responsibility to ensure that the correct files are submitted to Brightspace. No accommodation will be made for incorrect files or for submissions submitted to a different course. As unlimited submissions are permitted, students are strongly encouraged to submit early and verify their uploads to avoid submission errors. Only the final submission will be considered for grading.
- Students must achieve a 50% grade in the Theory component (Midterm and Final Exam – 20/40) and in the Practical component (Labs, Assignments & Project – 30/60) of the course separately to pass the course.
- Project report **AND** its presentation are **mandatory**.
- There will NOT be any makeup labs and assignments.
- The project report should be typed and saved in OneDrive so that, if needed, we can check the version history to see the contribution of each team member. OneDrive document should be downloaded and submitted in Brightspace before the due date.
- There will be in-class bonus quizzes at every lecture. If there are k quizzes, k-2 will be considered for bonus points. There will NOT be any make-up quizzes.

January 2026							February 2026							
Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	
					2	3		1	Lab 2 Demo	3	4	5	Lab 3 Due	7
4	5	6	7	8	9	10	8	Lab 3 Demo	10	11	12	Assignment 1 Due	14	
11	12	13	14	15	16	17	15	Family Day College Closed	17	18	19	Lab 4 Due	21	
18	19	20	21	22	Lab 1 Due	24	22	Assignment 1 Demo	24	25	26	27	28	
25	Lab 1 Demo	27	28	29	Lab 2 Due	31								
March 2026							April 2026							
Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	
1	Break	Break	Break	Break	Break	7				1	Final Project Due	Good Friday College Closed	4	
8	Lab 4 Demo	10	11	12	Lab 5 Due	14	5	Project Presentation	7	8	9	10	11	
15	Lab 5 Demo	17	18	19	Assignment 2 Due	21	12	Project Presentation	14	15	16	17	18	
22	Assignment 2 Demo	24	25	26	Project Part 1 Due	28	19	20	21	22	23	24	25	
29	Project Part 1 Demo	31					26	27	28	29	30			